

Markov chain
Monte Carlo

Recall our main problem of learning something about an unknown probability distribution f .

Sometimes, we have a feasible way of sampling directly from f . We can use this to produce many i.i.d samples.

Other times, we don't have an easy way to sample directly from f . In these situations, a method called **Markov chain Monte Carlo (MCMC)** can be useful.

The idea is to construct an ergodic Markov chain whose stationary distribution is f . We then simulate the chain for many steps to produce samples that are approximately from f .

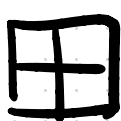
Consider an $n \times n$ checkerboard. Call a configuration of checkers on the checkerboard "valid" if no two checkers are adjacent (share a side). Assuming all valid configurations are equally likely, what is the average number of checkers in a valid configuration?

$$n=1$$



$$\text{Avg \# checkers} = \frac{1}{2}$$

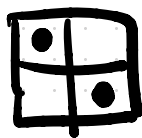
$$n=2$$



1



4



2

Avg # checkers

$$\frac{1}{7} \cdot 0 + \frac{4}{7} \cdot 1 + \frac{2}{7} \cdot 2 = \frac{8}{7}$$

Let $\mathcal{X}$ be the set of all valid configurations, and let f be the uniform distribution on $\mathcal{X}$. In other words,

$$f(x) = \frac{1}{|\mathcal{X}|} \quad \text{for all } x \in \mathcal{X}$$

We want the mean # of checkers in $\mathcal{X}$ according to distribution f ,

$$\sum_{x \in \mathcal{X}} \frac{1}{|\mathcal{X}|} (\# \text{ of checkers in } x)$$

If we could produce many iid samples from f , then we could use the mean number of checkers in the samples to estimate the true mean.

Method 1: Rejection sampling

Let $\mathcal{C}$ be the set of **all** checker configurations, valid or not.

Take a sample x from $\mathcal{C}$, uniformly at random.

- For every square, flip a coin. If heads, put a checker there.

i.e. $x \in \mathcal{X}$ is valid

If $x \in \mathcal{X}$, then accept the sample. If $x \notin \mathcal{X}$, reject the sample and try again. Keep doing this until a sample is accepted.

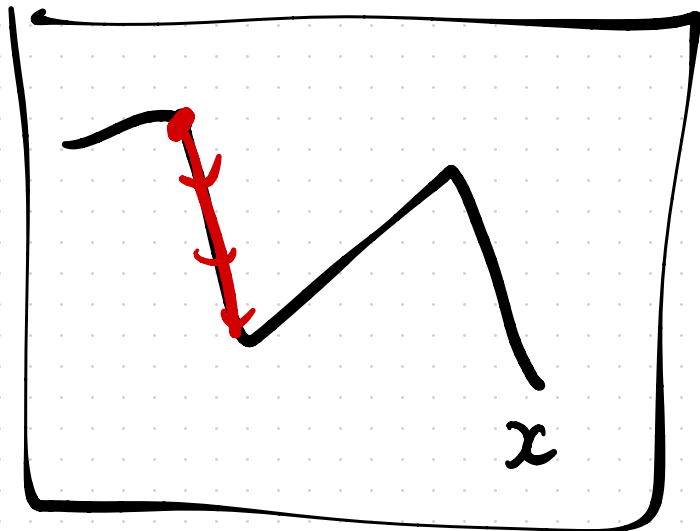
The accepted sample will be from $\mathcal{X}$, uniformly distributed.

This is a mathematically correct way of sampling from f .
It is known as **rejection sampling**.

This method would be useful if $\mathcal{X}$ was not much smaller than $\mathcal{C}$, so we would not reject too often.

However, in this example, $\mathcal{X}$ is much, much smaller than $\mathcal{C}$ for even moderately sized n . So it will take a long time to accept even a single sample.

⑥

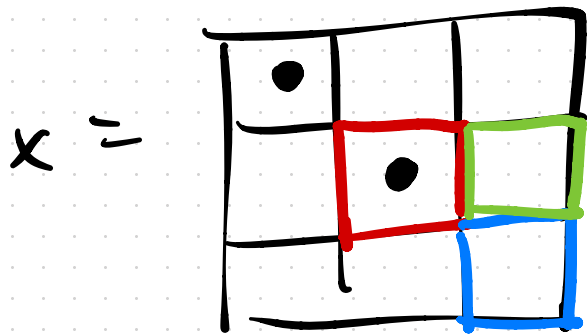


Method 2 : MCMC

Create a Markov chain as follows. The state space is $\mathcal{X}$.

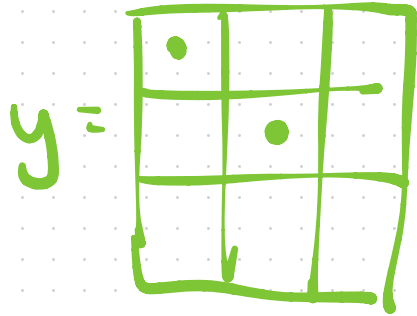
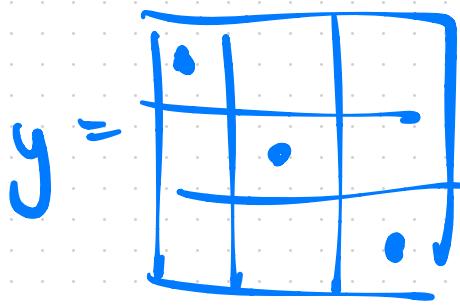
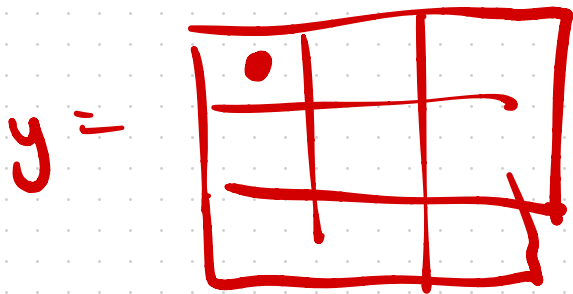
Suppose we are currently at state (configuration) x . Pick a square uniformly at random. If there is a checker there, remove it. Otherwise, place a checker there if the configuration remains valid; if not, stay at x this step.

Starting from a known valid configuration x_0 , ^{for example, empty config} we repeat the above process to get a Markov chain $x_0, x_1, x_2, \dots$. It can be shown that this chain is ergodic. What is the stationary distribution?



current configuration

Pick a square uniformly at random



Proposition (Detailed balance): Consider a Markov chain with transition probability matrix P . Suppose that π is some probability distribution such that for all states i and j ,

$$\pi_i P_{ij} = \pi_j P_{ji} \quad (*)$$

Then π is a stationary distribution of the Markov chain.

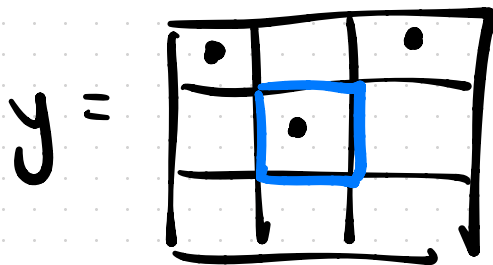
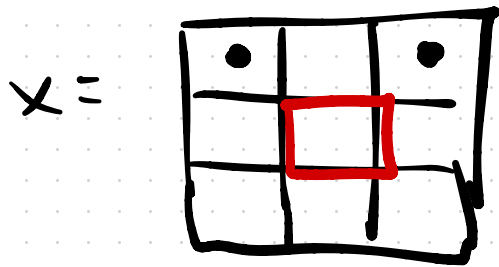
Warning: The converse is not true! Not all stationary distributions satisfy $(*)$.

Corollary: If P is symmetric ($P_{ij} = P_{ji}$ for all i, j), then the uniform distribution is a stationary distribution.

Consider the Markov chain on $\mathcal{X}$ from above, and let x and y be any two different states

If x and y differ in at least two squares, then $P_{xy} = 0$, $P_{yx} = 0$

If x and y differ in exactly one square



Then $P_{xy} = \frac{1}{n^2}$, because to transition from x to y we need to pick exactly the square where they differ.

By the same argument $P_{yx} = \frac{1}{n^2}$

So in every case, $P_{xy} = P_{yx}$
Thus P is symmetric, as the uniform distribution is stationary.

So our Markov chain is ergodic, and the uniform distribution is stationary.

Therefore, for large k , x_k will be approximately uniformly distributed on $\mathcal{X}$. We can estimate the average number of checkers of a valid configuration by simulating the Markov chain to obtain samples $x_1, x_2, \dots, x_N$ for large N , and finding the average number of checkers of the samples.

only need to record
of checkers at
each state

To avoid the first samples from skewing our results too much, we should have **burn-in period** where we run the Markov chain but don't collect data.